Why an answer score alone cannot diagnose document QA

Constructed example: four traces, one source document and one question

SOURCE: STAFFING REPORT

Category	2023	2024
Staff	**80**	**100**
Contractors	40	50

Note: staff exclude contractors.

QUESTION

How many staff were added in 2024?

Reference: 100 - 80 = 20 staff

No reason for the change is stated.

A numeric match can miss other errors.

A Correct number, wrong citation

Answer: 20 staff

Cites **Contractors row**

Trace: 100 - 80 = 20

The cited row does not supply these inputs.

B Correct cells, unsupported claim

Answer: 20 staff

"due to automation"

Trace: 100 - 80 = 20

Staff row cited; cause absent from source.

C Correct cells, arithmetic error

Answer: 30 staff

Cites **Staff row**

Trace: 100 - 80 = 30

Relevant evidence; incorrect calculation.

D All observed checks consistent

Answer: 20 staff

Cites **Staff row**

Trace: 100 - 80 = 20

Correct number with relevant support.

DIAGNOSTIC	Numeric match	Relevant cells	Cited support	Arithmetic (given inputs)
A Wrong citation	✓	✗	✗	✓
B Unsupported claim	✓	✓	✗	✓
C Arithmetic error	✗	✓	✗	✗
D Consistent trace	✓	✓	✓	✓

Causal faithfulness still needs evidence-removal or substitution controls.